Tutorial 4

MH1200 Linear Algebra I

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Ex. 6. Show that, for square matrices:
1. (a) $\operatorname{tr}(A+B) = \operatorname{tr}(A) + \operatorname{tr}(B)$, (b) $\operatorname{tr}(rA) = r\operatorname{tr}(A)$
2. $\operatorname{tr}(CD) = \operatorname{tr}(DC)$
3. $AB - BA \neq I$

Sol 6. Part 1a:

Let $A = \left(a_{i,j}\right)_{n\times n}$ and $B = \left(b_{i,j}\right)_{n\times n}$

Thus, $A + B = \left(a_{i,j} + b_{i,j}\right)_{n\times n}$

We thus have:

$$\begin{aligned}\operatorname{tr}(A+B) &= \textstyle\sum_{i=1}^{n}\left(a_{i,i} + b_{i,i}\right)\\ &= \textstyle\sum_{i=1}^{n}\left(a_{i,i}\right) + \sum_{i=1}^{n}\left(b_{i,i}\right)\\ &= \operatorname{tr}(A) + \operatorname{tr}(B)\end{aligned}$$

Part 1b:

Let $A = \left(a_{i,j}\right)_{n\times n}$ and $rA = \left(ra_{i,j}\right)_{n\times n}$. We thus have:

$$\begin{aligned}\operatorname{tr}(rA) &= \textstyle\sum_{i=1}^{n}\left(ra_{i,i}\right)\\ &= r\textstyle\sum_{i=1}^{n}\left(a_{i,i}\right)\\ &= r\operatorname{tr}(A)\end{aligned}$$

Part 2:

Let $C = \left(c_{i,j}\right)_{n\times n}$ and $D = \left(d_{i,j}\right)_{n\times n}$.

We thus have: $CD = \left(\sum_{j=1}^{n} c_{i,j}d_{j,i}\right)_{n\times n}$ and $DC = \left(\sum_{j=1}^{n} d_{i,j}c_{j,i}\right)_{n\times n}$.

Next, we note that:

$$\begin{aligned}\operatorname{tr}(CD) &= \textstyle\sum_{i=1}^{n}\sum_{j=1}^{n}\left(c_{i,j}d_{j,i}\right)\\ &= \textstyle\sum_{j=1}^{n}\sum_{i=1}^{n}\left(c_{i,j}d_{j,i}\right)\\ &= \textstyle\sum_{j=1}^{n}\sum_{i=1}^{n}\left(d_{j,i}c_{i,j}\right)\\ &= \operatorname{tr}(DC)\end{aligned}$$

Part 3:

Suppose that $AB - BA = I$

Thus,

$$\begin{aligned}\mathrm{tr}(AB-BA)&=\mathrm{tr}(AB)+\mathrm{tr}((-1)BA)\\&=\mathrm{tr}(AB)-\mathrm{tr}(BA)\\&=\mathrm{tr}(AB)-\mathrm{tr}(AB)\\&=0=\mathrm{tr}(I)\end{aligned}$$

Thus, $\mathrm{tr}(I)=0$, contradiction. Therefore,

$$AB-BA\neq I$$

Ex. 8. (a) Let A be an $n\times n$ matrix with the property that $A=-A^T$. Such a matrix is called **anti-symmetric**. Show that

$$\mathrm{tr}(A)=0$$

You may use any properties you know about trace if you wish.

(b) Let $A=\left(a_{i,j}\right)_{n\times n}$ and $B=\left(b_{i,j}\right)_{n\times n}$. Give a matrix X with the property that

$$\mathrm{tr}(XA)=\sum_{i=1}^{n}\sum_{j=1}^{n}a_{i,j}b_{i,j}$$

Sol 8. I claim that $X=B^T$ works.

Note that $B^T=\left(b_{j,i}\right)_{n\times n}$.

Notice that,

$$\begin{aligned}\mathrm{tr}(B^TA)&=\textstyle\sum_{i=1}^{n}\sum_{j=1}^{n}b_{i,j}^{T}a_{j,i}\\&=\textstyle\sum_{i=1}^{n}\sum_{j=1}^{n}b_{j,i}a_{j,i}\\&=\textstyle\sum_{i=1}^{n}\sum_{j=1}^{n}b_{i,j}a_{i,j}\end{aligned}$$

The expression produced by computing the trace is identical to the target expression, so indeed, $X=B^T$, as needed.

Ex. 9. Compute:

$$\begin{pmatrix}1&2\\-1&-1\end{pmatrix}^2$$

Then, use that to compute:

$$\begin{pmatrix}1&2&1&0\\-1&-1&0&1\\1&0&1&2\\0&1&-1&-1\end{pmatrix}^2$$

Sol 9. Let $A=\begin{pmatrix}1&2\\-1&-1\end{pmatrix}$ and $I=\begin{pmatrix}1&0\\0&1\end{pmatrix}$.

We note that:

$$A^2=\begin{pmatrix}1\cdot1+2\cdot(-1)&1\cdot2+2\cdot(-1)\\(-1)\cdot1+(-1)\cdot(-1)&(-1)\cdot2+(-1)\cdot(-1)\end{pmatrix}$$

$$=\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

Now, we have:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ -1 & -1 & 0 & 1 \\ 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & -1 \end{pmatrix}^2 = \begin{pmatrix} A & I \\ I & A \end{pmatrix}^2$$

$$= \begin{pmatrix} A^2+I^2 & AI+IA \\ AI+IA & I^2+A^2 \end{pmatrix}$$

$$= \begin{pmatrix} A^2+I & 2A \\ 2A & I+A^2 \end{pmatrix}$$

$$= \begin{pmatrix} \mathbf{0} & 2A \\ 2A & \mathbf{0} \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 & 2 & 4 \\ 0 & 0 & -2 & -2 \\ 2 & 4 & 0 & 0 \\ -2 & -2 & 0 & 0 \end{pmatrix}$$

Ex. 10. Let m and n be positive integers. Let $A, X \in \mathbb{M}_{m\times n}(\mathbb{R})$.

(a) Find an expression for $\mathrm{tr}(AX^T)$ in terms of the entries of the matrices, and (b) Show that $\forall A \in \mathbb{M}_{m\times n}(\mathbb{R})\ .\ \mathrm{tr}(AX^T) = 0 \to X = \mathbf{0}_{m\times n}$

Sol 10. Part (a):

Let $A = \left(a_{i,j}\right)$, $X = \left(x_{i,j}\right)$, $X^T = \left(x_{j,i}\right)_{n\times m}$

$$\mathrm{tr}(AX^T) = \sum_{i=1}^{n} \sum_{j=i}^{n} a_{i,j}x_{i,j}$$

Part (b):

Suppose that, for any matrix

$$A \in \mathbb{M}_{m\times n}(\mathbb{R})$$

,

$$\mathrm{tr}(AX^T) = \sum_{i=1}^{m} \sum_{j=i}^{n} a_{i,j}x_{i,j} = 0$$

We want to show $X = \mathbf{0}_{m\times n}$.

That is, we want to show:

$$\forall A \in \mathbb{M}_{m\times n}(\mathbb{R})\ .\ \mathrm{tr}(AX^T) = 0 \to X = \mathbf{0}_{m\times n}$$

We will instead prove the contrapositive, which is logically equivalent to the original statement:

$$X \neq \mathbf{0}_{m\times n} \to \neg\forall A \in \mathbb{M}_{m\times n}(\mathbb{R})\ .\ \mathrm{tr}(AX^T) = 0$$

$$\equiv$$

$$X \neq \mathbf{0}_{m\times n} \to \exists A \in \mathbb{M}_{m\times n}(\mathbb{R})\ .\ \mathrm{tr}(AX^T) \neq 0$$

Suppose that $X \in \mathbb{M}_{m\times n}(\mathbb{R})$ has a nonzero entry at row i_0, column j_0. That is, $x_{i_0,j_0} \neq 0$.

Then, let $A = \left(a_{i,j}\right)$, where

$$a_{i,j} = \begin{cases} 1 \text{ if } i = i_0 \wedge j = j_0 \\ 0 \text{ otherwise} \end{cases}$$

Notice that:

$$\begin{aligned} \operatorname{tr}(AX^T) &= \sum_{i=1}^{m} \sum_{j=1}^{n} a_{i,j} x_{i,j} \\ &= a_{i_0,j_0} x_{i_0,j_0} \\ &= x_{i_0,j_0} \\ &\neq 0 \end{aligned}$$

Thus, there exists a matrix A such that $\operatorname{tr}(AX^T) \neq 0$, as needed. ■

Ex. 11. Let A be a diagonal $n \times n$ matrix. Show that another matrix B exists such that $AB = I$ iff $a_{i,i} \neq 0$ for all $1 \leq i \leq n$

Sol 11. Let us first show the backward direction. Suppose

$$a_{i,i} \neq 0$$

Define

$$B = \left(b_{i,j}\right)_{n \times n}$$

, with:

$$b_{i,j} = \begin{cases} \frac{1}{a_{i,j}} \text{ if } i = j \\ 0 \text{ otherwise} \end{cases}$$

Then,

$$AB = \left(\sum_{k=1}^{n} a_{i,k} b_{k,j}\right)$$

We note that $\sum_{k=1}^{n} a_{i,k} b_{k,j}$ is nonzero only if there is some term where both $a_{i,k}$ and $b_{k,j}$ are nonzero. Thus, $\sum_{k=1}^{n} a_{i,k} b_{k,j}$ is nonzero only if $i = k \wedge k = j \rightarrow i = j$.

When $i = j$, we have:

$$\begin{aligned} \sum_{k=1}^{n} a_{i,k} b_{k,j} &= \sum_{k=1}^{n} a_{i,k} b_{k,i} \\ &= a_{i,i} b_{i,i} \\ &= a_{i,i} \cdot \frac{1}{a_{i,i}} \\ &= 1 \end{aligned}$$

Thus, AB consists of all zeros except at the diagonals. All diagonal entries are equal to 1. This implies that $AB = I$.

Now, we will show the forward direction. That is, if a matrix B exists so that $AB = I$, then $a_{i,i} \neq 0$ for all $1 \leq i \leq n$.

Note that the (i, i)-th entry of AB satisfies:

$$\sum_{k=1}^{n} a_{i,k} b_{k,i} = 1$$

Further, since $a_{i,j} = 0$ when $i \neq j$,

$$a_{i,i}b_{i,i} = 1$$

Note that

$$b_{i,i} \neq 0$$

as that would otherwise contradict $a_{i,i}b_{i,i} = 1$

Thus,

$$a_{i,i} = \frac{1}{b_{i,i}} \neq 0$$

As needed. ■

Ex. 14. We define the following terms:

Involutory Matrix A is involutory iff $A^2 = I$
Idempotent Matrix B is idempotent iff $B^2 = B$

Show that any involutory matrix can be expressed as the difference of two idempotent matrices.

Sol 14. Suppose A is an involutory matrix.

Notice that A must be a square matrix for A^2 to be defined. Therefore, $A + kI$ for any $k \in \mathbb{R}$ is also defined since A is square.

I claim that $\frac{A+I}{2}$ and $\frac{I-A}{2}$ are idempotent.

Notice that:

$$\begin{aligned}\left(\tfrac{A+I}{2}\right)^2 &= \tfrac{1}{4}(A^2 + AI + IA + I^2)\\ &= \tfrac{1}{4}(I + A + A + I)\\ &= \tfrac{A+I}{2}\end{aligned}$$

Also,

$$\begin{aligned}\left(\tfrac{I-A}{2}\right)^2 &= \tfrac{1}{4}(A^2 - AI - IA + I^2)\\ &= \tfrac{1}{4}(I - A - A + I)\\ &= \tfrac{I-A}{2}\end{aligned}$$

Thus, both $\frac{A+I}{2}$ and $\frac{I-A}{2}$ are idempotent.

Finally, notice that $A = \frac{A+I}{2} - \frac{I-A}{2}$.

Since $\frac{A+I}{2}$ and $\frac{I-A}{2}$ exist for any involutory matrix A, any involutory matrix can be expressed as the difference of two idempotent matrices, as needed. ■

Reflection on problem 14: The solution to this problem requires a clever construction that's similar to the proof that every function is the sum of an even and an odd function. When I noticed that $A + I$ and $I - A$ were almost involutory, I should've doubled down on that insight. What I could've noticed is that the equation in terms of $A + I$ is not homogenous, so cleverly using a scaling factor could've worked (Indeed it did).

Ex. 15. (i) Consider the matrix:

$$P = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Compute $P^3 - I_3$. Show your working.

(ii) Let a and b be real numbers. Calculate the following quantity:

$$\operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^9 \right)$$

(iii) Let a and b be real numbers. Calculate the following quantity:

$$\operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^{100} \right)$$

Sol 15. In the interest of time, I will only solve (iii). We show a generalization of this property for arbitrary positive integer exponents n.

Define a permutation matrix as a matrix where each row is composed of all 0s except for exactly one entry, which is 1. We note that $P^3 = I_3$. An easy way to see this is to consider the isomorphism between the group consisting of permutation matrices of order n under the operation of matrix multiplication and the symmetric group over n elements, S_n. A more elementary argument is presented here:

Since $P^3 = I_3$, I claim that the two matrices commute with one another.

Notice that,

$$\begin{aligned} I_3 P &= P^3 P \\ &= P^4 \\ &= P P^3 \\ &= P I_3 \end{aligned}$$

Since the matrices commute with each other, they satisfy the binomial theorem.

Thus, we have:

$$\begin{aligned} \operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n \right) &= \operatorname{tr}\left((aI_3 + bP)^n \right) \\ &= \operatorname{tr}\left((aP^3 + bP)^n \right) \\ &= \operatorname{tr}\left(\sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i} P^{3i} P^{n-i} \right) \\ &= \operatorname{tr}\left(\sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i} P^{n+2i} \right) \end{aligned}$$

Note that $\operatorname{tr}\left(P^{0 \bmod 3}\right) = 3$, and $\operatorname{tr}\left(P^{1 \bmod 3}\right) = \operatorname{tr}\left(P^{2 \bmod 3}\right) = 0$. Therefore,

$$\begin{aligned} \operatorname{tr}\left(\begin{pmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{pmatrix}^n \right) &= \operatorname{tr}\left(\sum_{i=0}^{n} \binom{n}{i} a^i b^{n-i} P^{n+2i} \right) \\ &= \operatorname{tr}\left(\sum_{i=0, i \equiv n \bmod 3}^{n} \binom{n}{i} a^i b^{n-i} P^{n+2i} \right) \end{aligned}$$

$$=3\sum_{j=0,j\equiv n \bmod 3}^{n}\binom{n}{j}a^j b^{n-j}$$

To simplify this sum, we will use generating functions and complex numbers. The appearance of the $n \bmod 3$ restriction motivates the introduction of complex numbers to handle the cycling behavior of modular arithmetic.

Let $P(x)=\sum_{j=0}^{n}\binom{n}{j}a^j b^{n-j}x^j=(ax+b)^n$ be the ordinary power series generating function describing our sequence. Further, let ζ_1,ζ_2,ζ_3 be the three roots of $x^3-1=0$ (i.e., the third roots of unity).

Notice that $\zeta_1^n+\zeta_2^n+\zeta_3^n=\begin{cases}0 \text{ if } n\not\equiv 0 \bmod 3\\ 3 \text{ if } n\equiv 0 \bmod 3\end{cases}$. Therefore, we have:

$$\begin{aligned}
\operatorname{tr}\left(\begin{pmatrix}a&0&b\\b&a&0\\0&b&a\end{pmatrix}^n\right)&=3\sum_{j=0,j\equiv n \bmod 3}^{n}\binom{n}{j}a^j b^{n-j}\\
&=\zeta_1^{2n}P(\zeta_1)+\zeta_2^{2n}P(\zeta_2)+\zeta_3^{2n}P(\zeta_3)\\
&=(a\zeta_1+b)^n+\zeta_2^{2n}(a\zeta_2+b)^n+\zeta_3^{2n}(a\zeta_3+b)^n\\
&=(a+b)^n+\left(\tfrac{-1+i\sqrt{3}}{2}\right)^{2n}\left(a\tfrac{-1+i\sqrt{3}}{2}+b\right)^n+\left(\tfrac{-1-i\sqrt{3}}{2}\right)^{2n}\left(a\tfrac{-1-i\sqrt{3}}{2}+b\right)^n\\
&=(a+b)^n+\left(a+\left(\tfrac{-1-i\sqrt{3}}{2}\right)b\right)^n+\left(a+\left(\tfrac{-1+i\sqrt{3}}{2}\right)b\right)^n
\end{aligned}$$

Thus, we have:

$$\operatorname{tr}\left(\begin{pmatrix}a&0&b\\b&a&0\\0&b&a\end{pmatrix}^n\right)=(a+b)^n+\left(a+\left(\frac{-1-i\sqrt{3}}{2}\right)b\right)^n+\left(a+\left(\frac{-1+i\sqrt{3}}{2}\right)b\right)^n$$

Where $i=\sqrt{-1}$.

Note to self: It's kinda crazy that complex numbers show up, but apparently the answer is related to a deeper fact about the trace of a matrix — that the trace of a matrix M is the sum of its eigenvalues.

Eigenvalues, in general, need not be real numbers. This is why we get complex numbers as the result.